LEARNER GUIDE

The rule behind the sequence

Quadratics. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

You have met a quadratic as a growing sequence (Lesson 1) and found its fingerprint, the constant second difference (Lesson 2). But where does the sequence come from? Every sequence is made by a rule — and the rule that makes a quadratic sequence is a quadratic expression, the very thing you will expand next lesson. This lesson joins those two faces. Look at the growing rectangle of dots for stage n: it has n rows and n + 1 columns, so its count is n(n + 1). Split it into an n by n square and one extra column, and the count is $n^2 + n$. So the sequence, the rule and the expression are all the same quadratic: $n(n + 1) = n^2 + n$. You will grow that rectangle and read the expression from it, substitute a rule to generate a whole sequence, check that a rule written as brackets and as an expanded expression give the same terms, and — as a stretch — see why the second difference is always 2a. (Positions are called n; when we study the general expression and its graph next, we will use x.) There is no calculator today. The question you are exploring is: how can a sequence and an expression be the same quadratic?

What you are learning to notice

- That every sequence is made by a rule, and the rule of a quadratic sequence is a quadratic expression.
- That the growing dot rectangle IS the expression: count = area = $n(n + 1) = n^2 + n$.
- That substituting the position n into the rule generates the sequence, and the second difference is 2 × the number in front of n^2.
- That a rule written as brackets and as an expanded expression are the same rule — turning one into the other is expanding.

Moving through it, one step at a time

In The dots are the expression, grow the rectangle and watch the count become $n^2 + n$ (a square plus one extra column). In Rule → Sequence machine, pick a rule and let it substitute n = 1…5 to build the sequence, with its difference table. In Two ways to write the rule, substitute a value into both a bracket form and an expanded form and see they match. In Why is the second difference 2a? (a stretch), see that the first differences form a linear sequence that climbs by 2a.

Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

Example 1 — build the sequence from a rule

The rule is n(n + 1). Substitute the position n = 1, 2, 3, 4, 5 to generate each term.

	+4		+6		+8		+10	
2		6		12		20		30

1. n = 1	1 × 2 = 2
2. n = 2	2 × 3 = 6
3. n = 3	3 × 4 = 12
4. n = 4	4 × 5 = 20

the sequence is 2, 6, 12, 20, 30

Example 2 — one rule, two forms

$(n + 1)(n + 2)$ and $n^2 + 3n + 2$ are the same rule. Substitute n = 1 into both to check they agree.

	+6		+8		+10		+12	
6		12		20		30		42

1. the brackets, n = 1	(2)(3) = 6
2. the expanded form, n = 1	1 + 3 + 2 = 6
3. both give the same term, so	one rule, two forms

$(n + 1)(n + 2) = n^2 + 3n + 2$

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (substituting a value into a rule (Lesson 1), the constant second difference (Lesson 2), and the area of a rectangle), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

NEO by Nudge Education · nudgeeducation.online · Curriculum by Gerry Docherty. Learner Guide v0.1.